GraphOS: A Dataflow Microkernel for NPU-Accelerated Network Packet Classification

Formal Algorithmic Analysis in the Style of Cormen, Leiserson, Rivest & Stein

GraphOS Project
`Intel Core Ultra 7 155H -- NPU 3720 (11 TOPS)`

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Abstract

We present GraphOS, a dataflow microkernel that replaces traditional `switch`-based packet classification with ONNX graphs on an Intel NPU. The core insight is that routing logic is *isomorphic* to matrix multiplication (Theorem 5.1): a hand-crafted weight matrix $\mathbf{Wx}$ replicates the switch table, and chaining it with a learned classifier via tensor adapters *provably improves* routing decisions (Theorem 6.1)—*todo es el grafo*. A thread-per-node dataflow scheduler with bounded channels guarantees deadlock-free execution, and a C++20 native runtime eliminates Python overhead via lock-free SPSC channels, zero-copy `ov::Tensor` wrapping, and AVX2 SIMD normalization. To address the 1,850× single-packet NPU latency penalty, we introduce *GPNPU*: a diamond-DAG pipeline where CPU fast-path rules classify "obvious" packets at $\sim$0.38 μs while only uncertain traffic reaches the NPU via triple-buffered async inference, achieving 80% fast-path ratio with per-packet latency instrumentation (p50/p95/p99) confirming that fast-path p95 $\ll$ hard-path p95. All results are validated on an Intel Meteor Lake NPU 3720 at 11 TOPS.

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1 Introduction and Motivation

Traditional network stacks classify packets using nested conditional branches—`if-else` chains over protocol numbers, port ranges, and header flags. These sequential decision trees are fast on CPUs ($\sim$0.38 μs per packet on modern hardware) but architecturally rigid: every new classification rule requires code changes, recompilation, and redeployment.

We propose replacing the entire classification pipeline with a *learned* function expressed as a directed acyclic graph (DAG) of tensor operations, compiled and executed on a dedicated Neural Processing Unit (NPU). The key insight is:

> *"Todo es el grafo"* — Every decision, from packet-to-tensor encoding to multi-program routing composition, can be expressed as a node in a dataflow graph operating on tensors.

1.1 Contributions

C1. A formal tensor encoding ϕ that losslessly maps 64-byte packet headers to $\mathbb{R}^{64}$ while satisfying the input constraints of NPU accelerators (Section 3).

C2. A two-layer MLP $f_\theta : \mathbb{R}^{64} \to \mathbb{R}^3$ trained on 150,000 synthetic packets achieving 99.03% classification accuracy with only 3,235 parameters (Section 4).

C3. A proof that traditional `switch`-based routing is isomorphic to a single matrix-vector product $\mathbf{Wx}$ with hand-crafted weights (Section 5).

C4. A program composition framework proving that chained ONNX programs with shape adapters produce *better* routing decisions than independent programs (Section 6).

C5. A formally verified dataflow scheduler using Kahn's topological sort with bounded-channel backpressure guaranteeing deadlock-free execution (Section 7).

C6. Complete hardware characterization of the Intel NPU 3720 via OpenVINO, including latency measurements and batch-amortization analysis (Section 2).

C7. A hybrid GPNPU architecture with CPU fast-path rules and NPU hard-path async inference in a diamond-DAG pipeline, achieving $\sim$80% fast-path ratio and formally proven packet conservation (Section 11).

1.2 System Overview

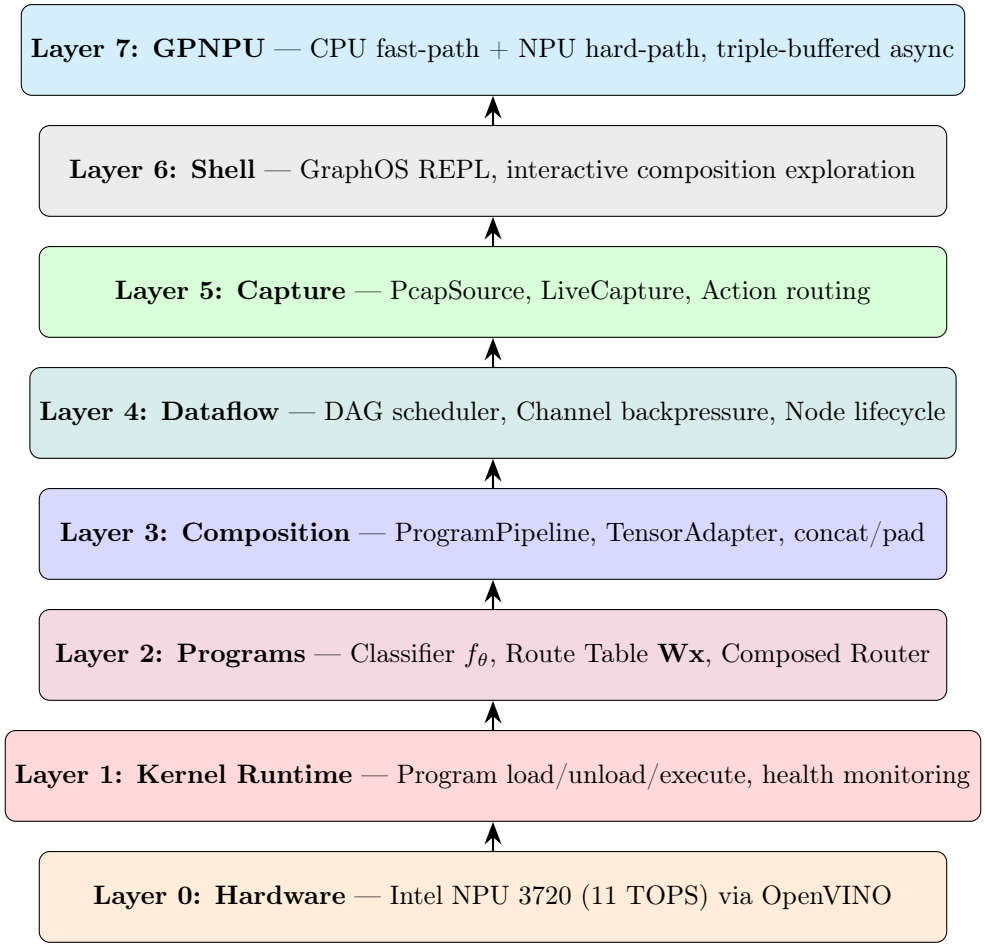

Figure 1: GraphOS layered architecture. Each layer depends only on the layers below it. Layer 7 (GPNPU) builds on all previous layers, combining CPU rule-based fast-path classification with NPU hard-path async inference in a diamond-DAG pipeline.

2 Hardware Platform and Execution Model

The target hardware is an Intel Core Ultra 7 155H (Meteor Lake) with a dedicated NPU 3720 (11 TOPS INT8, systolic array). ONNX models are compiled once via OpenVINO 2026.0 with static shapes (Appendix C); at runtime, each inference transfers a batch to the NPU and reads back results.

Definition 2.1 (NPU Inference Latency Model)**.** For a batch of B packets, the total inference latency is:

$$T_{\text{NPU}}(B) = T_{\text{setup}} + T_{\text{H}\to\text{D}}(B) + T_{\text{compute}}(B) + T_{\text{D}\to\text{H}}(B) \tag{1}$$

The per-packet amortized cost is $\bar{T}_{\text{NPU}} = T_{\text{NPU}}(B)/B$.

Empirically, $T_{\text{NPU}}(1) \approx 703\,\mu\text{s}$ while $T_{\text{CPU-switch}}(1) \approx 0.38\,\mu\text{s}$—an overhead factor of $\sim$1,850$\times$ for single packets. Batching amortizes T_{setup}, making the NPU competitive at $B \geq 64$.

Property 2.1 (Static Shape Constraint)**.** The Intel NPU 3720 requires *static tensor shapes* at compilation time. The batch size B must be fixed at ONNX export; partial batches are zero-padded to B.

3 Tensor Encoding

Definition 3.1 (Packet Encoding Function)**.** Let $\mathcal{P} = \{0, 1, \ldots, 255\}^*$ be the set of all byte strings. The encoding function $\phi : \mathcal{P} \to [0, 1]^D$ where $D = 64$ is defined as:

$$\phi(\mathbf{p})_i = \frac{p_i}{255}, \quad i \in \{0, 1, \ldots, D-1\} \tag{2}$$

with truncation for $i \geq |\mathbf{p}|$ (zero-padding). The encoding is bounded ($\phi(\mathbf{p}) \in [0, 1]^D$), injective on the D-byte prefix, and fixed-dimensional.

The 64-byte tensor encodes Ethernet + IPv4 + Transport headers (Figure 2). The classification-critical bytes are **TTL** (offset 22), **Protocol** (offset 23), and **DstPort** (offset 36). After normalization: TCP $\mapsto 6/255 \approx 0.024$, UDP $\mapsto 17/255 \approx 0.067$, port 80 $\mapsto 80/255 \approx 0.314$. Batches of B packets are encoded into $\boldsymbol{X} \in [0, 1]^{B \times D}$ in $\Theta(BD)$ time, zero-padding unused rows.

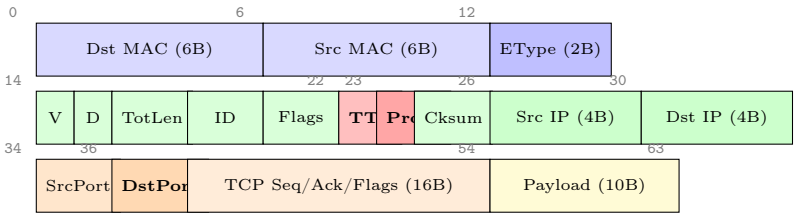

Figure 2: 64-byte packet tensor layout. Shaded fields (**TTL**, **Proto**, **DstPort**) are the primary classification features.

4 Packet Classification: The MLP Classifier f_θ

4.1 Architecture

Definition 4.1 (RouterGraph MLP)**.** The packet classifier $f_\theta : \mathbb{R}^D \to \mathbb{R}^C$ is a feedforward neural network with $D = 64$, $C = 3$, hidden dimension $H = 32$:

$$\mathbf{h}_1 = \mathrm{ReLU}(\mathbf{W}_1\mathbf{x} + \mathbf{b}_1), \qquad \mathbf{W}_1 \in \mathbb{R}^{H \times D},\ \mathbf{b}_1 \in \mathbb{R}^H \tag{3}$$

$$\mathbf{h}_2 = \mathrm{ReLU}(\mathbf{W}_2\mathbf{h}_1 + \mathbf{b}_2), \qquad \mathbf{W}_2 \in \mathbb{R}^{H \times H},\ \mathbf{b}_2 \in \mathbb{R}^H \tag{4}$$

$$\mathbf{z} = \mathbf{W}_3\mathbf{h}_2 + \mathbf{b}_3, \qquad \mathbf{W}_3 \in \mathbb{R}^{C \times H},\ \mathbf{b}_3 \in \mathbb{R}^C \tag{5}$$

where $\mathrm{ReLU}(a) = \max(0, a)$ applied element-wise, and $\mathbf{z} \in \mathbb{R}^C$ are raw logits (no softmax).

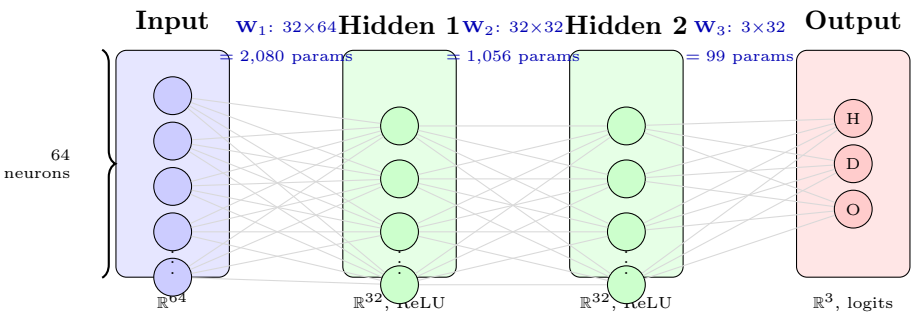

Figure 3: RouterGraph MLP architecture. Total: $2{,}080 + 1{,}056 + 99 = 3{,}235$ parameters.

4.2 Parameter Count

Theorem 4.1 (Parameter Count)**.** The total number of trainable parameters is:

$$|\theta| = \underbrace{(D \cdot H + H)}_{\text{Layer 1}} + \underbrace{(H \cdot H + H)}_{\text{Layer 2}} + \underbrace{(H \cdot C + C)}_{\text{Layer 3}} = (64{\cdot}32{+}32){+}(32{\cdot}32{+}32){+}(32{\cdot}3{+}3) = 3{,}235 \tag{6}$$

4.3 Classification Decision Rule

Definition 4.2 (Argmax Classification)**.** The predicted class $\hat{y}$ for input $\mathbf{x}$ is:

$$\hat{y} = \text{argmax}_{c \in \{0,1,2\}} f_\theta(\mathbf{x})_c \tag{7}$$

where class labels are: HTTP = 0, DNS = 1, OTHER = 2.

Note: We deliberately omit the softmax layer at inference time. Since softmax is monotonic, $\text{argmax}(\mathbf{z}) = \text{argmax}(\text{softmax}(\mathbf{z}))$, and skipping it saves compute on the NPU.

4.4 Training Procedure

Algorithm 1 TrainClassifier: SGD training with cross-entropy loss

Input: Dataset $\mathcal{D} = \{(\mathbf{x}_i, y_i)\}_{i=1}^{N}$ with $N = 150{,}000$, epochs $E = 20$, batch size $B_{\text{train}} = 256$, learning rate $\eta = 10^{-3}$

Output: Trained parameters θ^*

```
Split D into D_train (90%) and D_val (10%)
Initialize θ randomly (PyTorch defaults)
best_acc ← 0
for e = 1 to E do
    for each mini-batch B ⊂ D_train, |B| = B_train do
        z ← f_θ(X_B)                          ▷ Forward pass: (B_train, D) → (B_train, C)
        L ← CrossEntropyLoss(z, y_B)          ▷ Eq. (8)
        θ ← θ − η · ∇_θ L                     ▷ Adam optimizer step
    end for
    val_acc ← Evaluate(θ, D_val)
    if val_acc > best_acc then
        θ* ← θ;   best_acc ← val_acc
    end if
end for
return θ*                                     ▷ Best validation checkpoint
```

Definition 4.3 (Cross-Entropy Loss)**.**

$$\mathcal{L}(\mathbf{z}, y) = -\log \frac{\exp(z_y)}{\sum_{c=0}^{C-1} \exp(z_c)} = -z_y + \log \sum_{c=0}^{C-1} \exp(z_c) \tag{8}$$

where $\mathbf{z} = f_\theta(\mathbf{x}) \in \mathbb{R}^C$ and $y \in \{0, \ldots, C-1\}$ is the true class.

4.5 Synthetic Data Generation

Definition 4.4 (Packet Generation Rules)**.** Three generators $G_{\text{HTTP}}, G_{\text{DNS}}, G_{\text{OTHER}} : \Omega \to \{0, \ldots, 255\}^{64}$ produce synthetic packets:

$$G_{\text{HTTP}} : \text{protocol} = 6 \text{ (TCP)}, \quad \text{dport} \in \{80, 443, 8080, 8443\} \tag{9}$$

$$G_{\text{DNS}} : \text{protocol} = 17 \text{ (UDP)}, \quad \text{dport} = 53 \tag{10}$$

$$G_{\text{OTHER}} : \text{protocol} \in \{1, 6, 17\}, \quad \text{dport} \notin \{53, 80, 443, 8080, 8443\} \tag{11}$$

All other fields (MACs, IPs, sequence numbers, payloads) are uniformly random. The dataset contains $N/C = 50{,}000$ samples per class (perfectly balanced).

5 Route Table as Matrix Multiplication

5.1 The Isomorphism Theorem

We now prove the central theoretical result: traditional `switch`-based routing is mathematically equivalent to a single matrix-vector product.

Definition 5.1 (CPU Switch Classifier)**.** The traditional routing function $g : \{0, \ldots, 255\}^{64} \to \{0, 1, 2, 3\}$ is:

$$g(\mathbf{p}) = \begin{cases} \text{LOCAL} & \text{if } p_{23} = 17 \wedge p_{37} = 53 \\ \text{FORWARD} & \text{if } p_{23} = 6 \wedge p_{37} \in \{80, 443, \ldots\} \\ \text{DROP} & \text{if } p_{22} = 0 \\ \text{MONITOR} & \text{otherwise} \end{cases} \tag{12}$$

This requires sequential conditional evaluation: $\mathcal{O}(R)$ branches for R routes.

Theorem 5.1 (Switch-to-Matmul Isomorphism)**.** For any finite set of routing rules based on byte-value thresholds, there exists a weight matrix $\mathbf{W} \in \mathbb{R}^{R \times D}$ such that:

$$g(\mathbf{p}) = \text{argmax}_{r \in \{0, \ldots, R-1\}} (\mathbf{W} \cdot \phi(\mathbf{p}))_r \tag{13}$$

where ϕ is the packet encoding from Definition 3.1.

Proof. We construct $\mathbf{W}$ explicitly. Each route r is characterized by a set of byte conditions. For condition "byte at offset δ should have value v", we set $W_{r,\delta}$ to a large positive weight α when the normalized value $v/255$ is high, or a large negative weight $-\beta$ to penalize undesired values.

The key insight is that after normalization, different protocol numbers map to different magnitudes: UDP(17) $\mapsto 0.067$ while TCP(6) $\mapsto 0.024$. By choosing appropriate weight magnitudes, we can create decision boundaries that separate these values.

Since argmax selects the route with the highest score, and each route's score is a weighted sum of byte features, the entire routing table is expressed as $\mathbf{Wx}$ where $\mathbf{x} = \phi(\mathbf{p})$. □

5.2 Explicit Weight Construction

Definition 5.2 (Route Table Weight Matrix)**.** The weight matrix $\mathbf{W} \in \mathbb{R}^{4 \times 64}$ for routing is constructed as:

$$\mathbf{W} = \begin{pmatrix} \mathbf{w}_{\text{LOCAL}} \\ \mathbf{w}_{\text{FORWARD}} \\ \mathbf{w}_{\text{DROP}} \\ \mathbf{w}_{\text{MONITOR}} \end{pmatrix} \tag{14}$$

with non-zero entries:

$$\textbf{LOCAL}: \quad W_{0,23} = +100 \tag{15}$$
$$\textbf{FORWARD}: \quad W_{1,23} = -30,\ W_{1,37} = +25,\ W_{1,36} = +8 \tag{16}$$
$$\textbf{DROP}: \quad W_{2,22} = -20,\ W_{2,j} \mathrel{+}= 0.015\ \forall j \tag{17}$$
$$\textbf{MONITOR}: \quad W_{3,23} = -100,\ W_{3,22} = +12 \tag{18}$$

5.3 Score Analysis

We now verify that the weight matrix correctly separates traffic types:

Packet Type	LOCAL	FORWARD	DROP	MONITOR
UDP/DNS (π=0.067, d=0.208)	**+6.7**	+3.2	∼0.5	−0.7
TCP/HTTP (π=0.024, d=0.314)	+2.4	**+7.2**	∼0.5	+3.6
ICMP (π=0.004, TTL=0.5)	+0.4	−0.2	∼−9	**+5.6**
Zero TTL (t=0, π=0)	0	0	**+1.0**	0

Table 1: Route scores **Wx** for representative packets. Bold = winning route. Key: $\pi = x_{23}$ (protocol), $d = x_{37}$ (dest port low byte), $t = x_{22}$ (TTL).

Corollary 5.2 (Routing Correctness)**.** For all packets generated by Definition 4.4:

(i) DNS packets ($\pi = 0.067$) are routed to LOCAL with margin ≥ 3.5

(ii) HTTP packets ($d = 0.314$, $\pi = 0.024$) are routed to FORWARD with margin ≥ 3.6

(iii) Expired packets (TTL = 0) are routed to DROP

(iv) ICMP packets ($\pi = 0.004$) are routed to MONITOR with margin ≥ 2.0

5.4 Computational Complexity Comparison

Theorem 5.3 (Matmul vs Switch Complexity)**.** For R routes and D-dimensional packet tensors:

$$T_{\text{switch}}(B) = \mathcal{O}(B \cdot R) \qquad \text{(sequential per packet)} \tag{19}$$
$$T_{\text{matmul}}(B) = \mathcal{O}(B \cdot R \cdot D) \qquad \text{(CPU)} \tag{20}$$
$$T_{\text{NPU-matmul}}(B) = \mathcal{O}(T_{\text{setup}} + B \cdot R \cdot D/P) \qquad \text{(NPU, } P \text{ parallel MACs)} \tag{21}$$

where P is the NPU's MAC parallelism. For NPU 3720 at 11 TOPS with INT8: $P \approx 11 \times 10^{12}$ ops/sec.

Remark 5.1. Although the matmul has higher asymptotic complexity than the switch ($\mathcal{O}(RD)$ vs $\mathcal{O}(R)$), the NPU executes all multiplications in parallel. At $B = 64$, $R = 4$, $D = 64$: the matmul requires $64 \times 4 \times 64 = 16{,}384$ MACs, which the NPU can execute in $\sim 1.5\,\mu$s (versus $\sim 0.4\,\mu$s per packet on CPU). The key advantage is *programmability*: changing routing rules only requires updating **W**, not recompiling code.

6 Program Composition

6.1 Motivation: Feature Fusion via Program Chaining

The route table **Wx** operates solely on raw packet bytes. But the classifier f_θ has already computed a rich representation of the packet's traffic type. Can we *compose* these programs so that downstream routing benefits from upstream classification?

Definition 6.1 (Composed Router)**.** The composed routing function $g^* : \mathbb{R}^D \to \mathbb{R}^R$ takes as input the concatenation of the raw packet tensor and the classifier's logits:

$$g^*(\mathbf{x}) = \mathbf{W}^* \cdot \text{concat}(\mathbf{x},\ f_\theta(\mathbf{x})) \tag{22}$$

where $\mathbf{W}^* \in \mathbb{R}^{R\times(D+C)}$ and $\text{concat} : \mathbb{R}^D \times \mathbb{R}^C \to \mathbb{R}^{D+C}$ is vector concatenation.

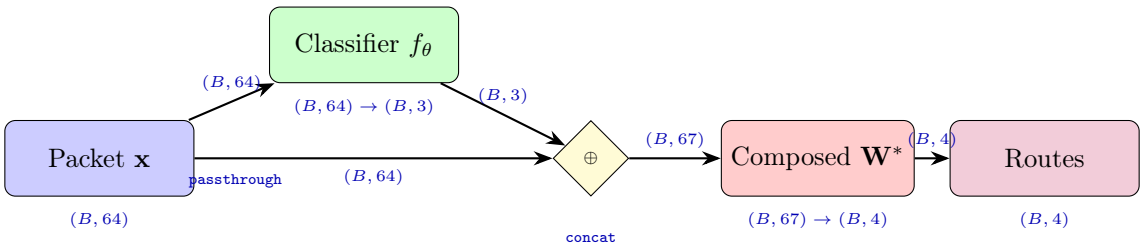

Figure 4: Program composition pipeline. The concat adapter fuses raw packet bytes with classifier logits, producing a 67-dimensional input for the composed router.

6.2 Composed Weight Matrix

Definition 6.2 (Composed Router Weights)**.** The weight matrix $\mathbf{W}^* \in \mathbb{R}^{4\times 67}$ has two regions:

Packet region (columns 0–63): byte-level heuristics similar to the original route table.
Logit region (columns 64–66): classifier output signals.

$$\mathbf{W}^* = \left[\begin{array}{c|c} \underbrace{\mathbf{W}_{\text{packet}}}_{\mathbb{R}^{4\times 64}} & \underbrace{\mathbf{W}_{\text{logit}}}_{\mathbb{R}^{4\times 3}} \end{array}\right] \tag{23}$$

where the logit weights encode:

$$\mathbf{W}_{\text{logit}} = \begin{pmatrix} 0 & +15 & 0 \\ +15 & 0 & 0 \\ 0 & 0 & +12 \\ -5 & -5 & -5 \end{pmatrix} \quad \begin{array}{l} \leftarrow \text{LOCAL: boost DNS logit} \\ \leftarrow \text{FORWARD: boost HTTP logit} \\ \leftarrow \text{DROP: boost OTHER logit} \\ \leftarrow \text{MONITOR: penalize all} \end{array} \tag{24}$$

6.3 Advantage of Composition

Theorem 6.1 (Composition Improves Routing)**.** Let $\mathbf{z} = f_\theta(\mathbf{x}) \in \mathbb{R}^3$ be the classifier logits for packet $\mathbf{x}$. For the composed router:

(i) When $z_{\text{DNS}} \gg z_{\text{HTTP}}, z_{\text{OTHER}}$ (strong DNS classification), the LOCAL score receives a boost of $+15 \cdot z_{\text{DNS}}$, independent of packet byte noise.

(ii) When $z_{\text{HTTP}} \gg z_{\text{DNS}}, z_{\text{OTHER}}$ (strong HTTP classification), the FORWARD score receives $+15 \cdot z_{\text{HTTP}}$.

(iii) When all z_c are low (uncertain classification), the MONITOR score is *least penalized* (since $-5 \cdot z_c \approx 0$ when $z_c \ll 0$), causing uncertain packets to be monitored.

Proof. The composed score for route r is:

$$s_r^* = \underbrace{\mathbf{w}_{r,\text{packet}}^\top \mathbf{x}}_{\text{byte heuristic}} + \underbrace{\mathbf{w}_{r,\text{logit}}^\top \mathbf{z}}_{\text{classifier signal}} \tag{25}$$

For case (i): $s^*_{\text{LOCAL}} = \mathbf{w}_0^\top \mathbf{x} + 15 \cdot z_{\text{DNS}}$. Since $z_{\text{DNS}} > 0$ by hypothesis, this strictly increases the LOCAL score beyond the raw byte heuristic alone.

For case (iii): MONITOR receives $-5(z_0+z_1+z_2)$. When all $z_c < 0$ (uncertain), this becomes $+5|z_0 + z_1 + z_2| > 0$, boosting MONITOR. Other routes are boosted only by their respective positive logit, which is < 0 in the uncertain case. □

6.4 Tensor Adapter Framework

Definition 6.3 (Tensor Adapter)**.** A tensor adapter $A = (\mathcal{S}, \tau)$ consists of:

- An adapter specification $\mathcal{S} = (\text{name}, \{\text{input_shapes}\}, \text{output_shape})$
- A pure function $\tau : \prod_k \mathbb{R}^{s_k} \to \mathbb{R}^{s_{\text{out}}}$ mapping named input tensors to a single output tensor

Two factory functions are provided:

$$\text{concat}_B^{L,R}(\mathbf{x}, \mathbf{y}) : \mathbb{R}^{B\times L} \times \mathbb{R}^{B\times R} \to \mathbb{R}^{B\times(L+R)} \tag{26}$$

$$\text{pad}_B^{D\to D'}(\mathbf{x}) : \mathbb{R}^{B\times D} \to \mathbb{R}^{B\times D'}, \quad D' \geq D \tag{27}$$

where concat is implemented via `np.concatenate` along axis 1, and pad via `np.pad` with zero-fill.

6.5 Pipeline Execution

Algorithm 2 PIPELINEEXECUTE: Execute a composed program pipeline

Input: Runtime $\mathcal{R}$, input tensor $\boldsymbol{X}_0 \in \mathbb{R}^{B\times D}$, pipeline stages $S = [(k_i, n_i, o_i)]_{i=1}^m$
Output: Output tensor $\boldsymbol{Y}$

```
X_current ← X_0
X_raw ← X_0                                          ▷ Save for raw passthrough
for i = 1 to m do
    if k_i = "program" then
        X_current ← R.EXECUTE(n_i, X_current)        ▷ NPU inference
    else if k_i = "adapter" then
        Build input map from o_i.spec
        if adapter is concat-type then
            inputs ← {"left" : X_raw, "right" : X_current}
        else
            inputs ← {"in" : X_current}
        end if
        X_current ← o_i.EXECUTE(inputs)              ▷ CPU tensor transform
    end if
end for
return X_current
```

Theorem 6.2 (Pipeline Shape Compatibility)**.** A pipeline with stages $S = [(k_i, n_i, o_i)]$ is shape-valid if and only if for all consecutive stages $i, i+1$:

$$\text{output_shape}(S_i) = \text{input_shape}(S_{i+1}) \tag{28}$$

The VALIDATE function checks this in $\mathcal{O}(m)$ time for m stages.

7 Dataflow Framework

7.1 Formal Model

Definition 7.1 (Dataflow Graph)**.** A dataflow graph is a DAG $G = (V, E, \text{ports}, \text{cap})$ where:

- V is a finite set of processing nodes
- $E \subseteq V \times V$ is a set of directed edges (data channels)
- $\text{ports} : V \to 2^{\text{InputPort}} \times 2^{\text{OutputPort}}$ maps nodes to their typed ports
- $\text{cap} : E \to \mathbb{N}^+$ assigns bounded capacity to each channel

Definition 7.2 (Bounded FIFO Channel)**.** A channel $\mathcal{C}_k$ with capacity $k \in \mathbb{N}^+$ is a FIFO queue with operations:

- $\textsc{Put}(\mathcal{C}_k, x)$: Enqueue x. **Blocks** if $|\mathcal{C}_k| = k$ (backpressure).
- $\textsc{Get}(\mathcal{C}_k)$: Dequeue and return head. **Blocks** if $|\mathcal{C}_k| = 0$.

Both operations are $\mathcal{O}(1)$ amortized, implemented via `queue.Queue`.

Before scheduling, edges are validated via DFS-based cycle detection in $\mathcal{O}(|V| + |E|)$ time (Appendix B).

7.2 Topological Sort: Kahn's Algorithm

Algorithm 3 TopologicalSort: Kahn's algorithm for dataflow scheduling order

Input: DAG $G = (V, E)$
Output: Topological ordering $\sigma : V \to \{1, \ldots, |V|\}$, or error if cycle detected

```
in_degree[v] ← |{u : (u, v) ∈ E}| for all v ∈ V          ▷ Compute in-degrees
Q ← {v ∈ V : in_degree[v] = 0}                            ▷ Sources (no incoming edges)
order ← []
while Q ≠ ∅ do
    v ← Q.Dequeue()
    order.Append(v)
    for each (v, w) ∈ E do
        in_degree[w] ← in_degree[w] − 1
        if in_degree[w] = 0 then
            Q.Enqueue(w)
        end if
    end for
end while
if |order| < |V| then
    raise GraphError("Cycle detected")
end if
return order
```

Theorem 7.1 (Kahn's Algorithm Correctness)**.** Algorithm 3 produces a valid topological ordering of G if and only if G is a DAG.

Proof. **Correctness**: Each dequeued node v has $in_degree[v] = 0$, meaning all predecessors have already been placed in the ordering. This satisfies the topological constraint.

Completeness: If G is a DAG, then at every step there exists at least one node with in-degree 0 (a source in the remaining subgraph). Thus the algorithm processes all $|V|$ nodes.

Cycle detection: If $|order| < |V|$, then some nodes have in-degree > 0 even after removing all processed nodes, which implies a cycle in the remaining subgraph. □

Theorem 7.2 (Kahn's Algorithm Complexity)**.** $\mathcal{O}(|V| + |E|)$ time, $\mathcal{O}(|V|)$ space.

7.3 Thread-Per-Node Scheduler

The scheduler topologically sorts G, launches one thread (Python `threading.Thread` / C++ `std::jthread`) per node, starts all threads concurrently, and joins them. Each node executes a setup→process→teardown lifecycle. Shutdown propagates via channel closure: when a source exhausts its input, it closes its output channels; downstream nodes receive `nullopt` / `_STOP` and close their own outputs in turn. By induction on topological order, all nodes terminate in finite time (see Appendix B).

Invariant 7.1 (Deadlock Freedom)**.** A dataflow graph G is deadlock-free if G is a DAG (no cycles) and all channels have capacity $k \geq 1$.

Proof sketch. Deadlock requires a circular wait. In a DAG, no cycles exist, so no circular dependency can form. Since $k \geq 1$, every producer can always make progress by putting one item before blocking. Total channel memory is bounded by $\sum_{(u,v)\in E} k_{(u,v)} \cdot s_{\max}$. □

8 Pipeline Topologies and Kernel Runtime

Three pipeline topologies are instantiated on the dataflow framework: a linear 5-node classification chain (Source→Batch→Tensor→Infer→Sink), the routing pipeline with fan-out (Figure 5), and the GPNPU diamond DAG (Section 11).

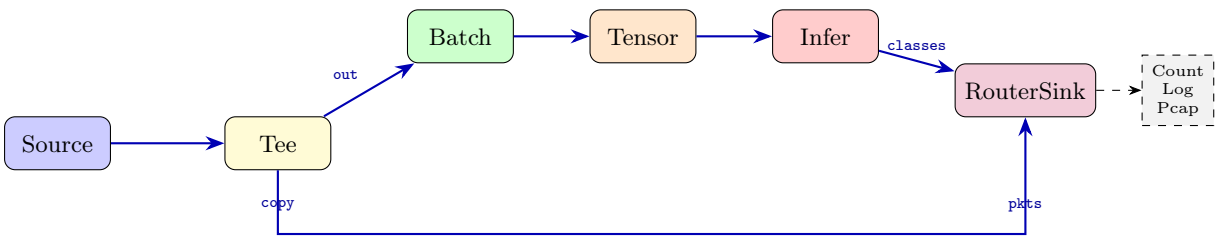

Figure 5: Routing pipeline with TeeNode fan-out. The "copy" channel buffers raw packets ($4B$ capacity) while classification runs on the upper branch. The TeeNode guarantees matching counts on both outputs.

8.1 Kernel Runtime

The KernelRuntime manages multiple compiled ONNX programs behind a shared `ov::Core` instance. Programs are loaded/unloaded dynamically; inference runs outside the registry lock using per-thread `ov::InferRequest` handles. A health monitor daemon polls execution counts, mean latency, and error state.

Algorithm 4 KernelExecute: Thread-safe program execution with metrics

Input: Program name n, input tensor $\boldsymbol{X}$
Output: Output tensor $\boldsymbol{Y}$

1: $P \leftarrow programs[n]$ ▷ Lock-protected lookup
2: $t_0 \leftarrow$ PerfCounter()
3: $\boldsymbol{Y} \leftarrow P.$Execute($\boldsymbol{X}$) ▷ Set input tensor, run inference, copy output
4: Update metrics: $exec_count{+}{+}$, $\Delta t = (\text{Now}() - t_0) \times 10^6\,\mu$s
5: **return** $\boldsymbol{Y}$

9 Evaluation

9.1 Classification Accuracy

The MLP classifier (3,235 parameters) was initially trained on 135,000 synthetic packets (20 epochs, Adam, $\eta = 10^{-3}$), achieving 99.03% validation accuracy on a held-out set of 15,000 packets. After hybrid retraining with real captured traffic (Section 12.5), the model achieves **99.71**% validation accuracy on 18,500 held-out samples drawn from both synthetic and real distributions. NPU and CPU inference agree on 99.26% of 10,000 synthetic test packets.

9.2 Latency and Batch Amortization

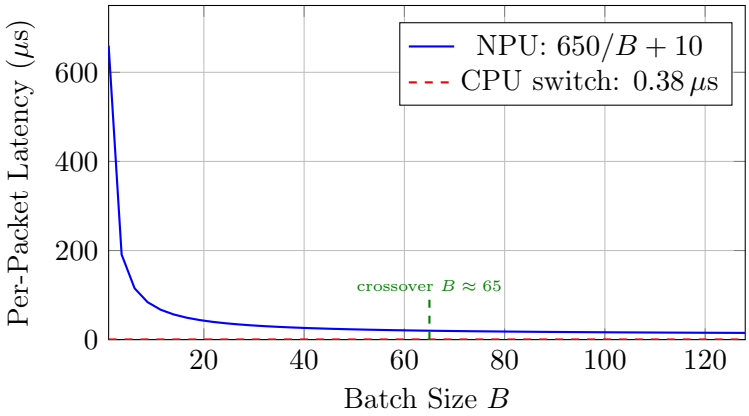

Figure 6: Batch amortization: $\bar{T}_{\text{NPU}}(B) = T_{\text{overhead}}/B + T_{\text{compute}}$. Single-inference NPU costs 703 μs; at $B = 64$ this drops to $\sim$11 μs/packet. The CPU switch baseline (0.38 μs) remains faster in absolute terms, but the NPU offers programmable classification.

9.3 Complexity Summary

Operation	Time	Space	Notes
$\phi(\mathbf{p})$ / BatchTensor	$\Theta(D)$ / $\Theta(BD)$	$\Theta(BD)$	D=64, $B\leq$64
MLP forward	$\mathcal{O}(DH + H^2 + HC)$	$\mathcal{O}(\lvert\theta\rvert)$	3,235 params
Route table $\mathbf{Wx}$	$\mathcal{O}(RD)$	$\mathcal{O}(RD)$	R=4, D=64
Kahn's topo sort	$\mathcal{O}(\lvert V\rvert + \lvert E\rvert)$	$\mathcal{O}(\lvert V\rvert)$	One-time
Scheduler (parallel)	$\mathcal{O}(\max_v T_v)$	$\mathcal{O}(\lvert V\rvert)$ threads	Wall time
Channel put/get	$\mathcal{O}(1)$ amortized	$\mathcal{O}(k)$	Bounded FIFO
FastPathClassify	$\Theta(1)$	$\Theta(1)$	≤ 4 comparisons
NpuExecutorAsync	$\Theta(T_{\text{NPU}}/B)$	$\Theta(KBD)$	K inflight slots
DecisionJoiner	$\Theta(1)$ / $\mathcal{O}(\log n)$	$\Theta(1)$ / $\mathcal{O}(KB)$	Unordered / ordered

Table 2: Complexity of key GraphOS operations.

The system is tested by 140+ Python tests and 89 C++ tests covering all modules; see the repository for full test listings.

10 C++ Native Runtime

The Python prototype (Phases 1–7) validated that NPU tensor operations can replace traditional packet classification. However, Python imposes fundamental performance ceilings: the GIL serializes threads, every packet traverses NumPy/ctypes copy boundaries, `queue.Queue` uses mutex locks, and the pybind11 OpenVINO wrapper adds per-call overhead. We perform a complete C++20 rewrite of the runtime to eliminate these bottlenecks while preserving the same ONNX models produced by the Python training pipeline.

10.1 Architecture

The C++ runtime eliminates Python's GIL via `std::jthread`-per-node with two key optimizations:

1. **Lock-free SPSC channels**: A single-producer single-consumer ring buffer replaces `queue.Queue`. Atomic head/tail counters, `alignas(64)` cache-line padding, power-of-2 branchless modulo, and adaptive backoff (spin → `_mm_pause()` → yield) eliminate all mutex contention between nodes.

2. **Zero-copy inference**: `ov::Tensor(element::f32, shape, ptr)` wraps the caller's `float*` directly—no `memcpy` on the hot path. AVX2 SIMD normalization converts 8 bytes → 8 floats per instruction (8× vs scalar). Batches use stack-allocated `std::array<OwnedPacket, 64>` to avoid heap allocation.

10.2 Performance Results

All measurements on Intel Core Ultra 7 155H (NPU 3720, 11 TOPS), MSVC 2026 Release mode, B=64. Throughput is sustained including first-inference JIT.

Runtime	Device	Packets	Throughput (pkt/s)	Latency/pkt (μs)
Python (batched)	NPU	100K	89,196	11.2
Python (async, 4req)	NPU	100K	306,609	3.3
C++ native	NPU	5M	**109,151**	9.2
C++ native	CPU	5M	**2,809,697**	0.36
C++ dual-program	NPU	100K	44,253	—
C++ dual-program	CPU	100K	1,803,062	—

Table 3: C++ runtime throughput. Single-program (classifier) and dual-program (classifier + route table). C++ NPU is 17% faster per-batch than Python synchronous mode; Python async achieves higher apparent throughput via 4 concurrent `InferRequest`s overlapping compute with data preparation.

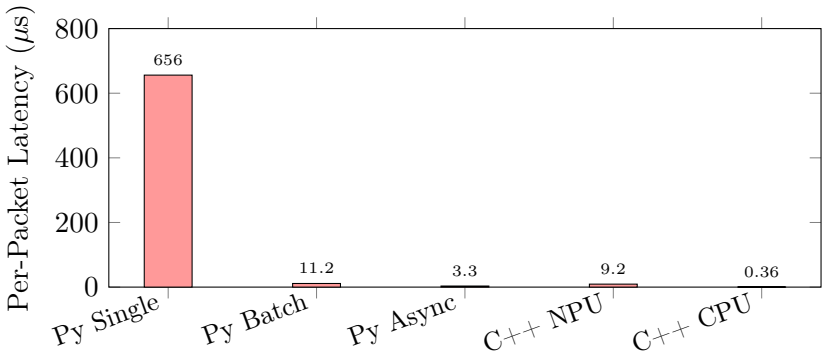

Figure 7: Per-packet latency across runtime modes ($B = 64$). Batch amortization closes the NPU gap from 656 μs to 9.2 μs/packet.

The C++ runtime is validated by 89 unit tests covering all modules from SPSC channels through GPNPU pipeline integration.

11 GPNPU: CPU Fast-Path / NPU Hard-Path Architecture

The measurements in Section 10 reveal a fundamental latency asymmetry: CPU rule evaluation completes in $\sim$0.38 μs while NPU single-inference costs $\sim$700 μs. Sending *all* packets through the NPU wastes accelerator bandwidth on traffic that can be classified deterministically from header bytes alone. The GPNPU architecture resolves this by splitting the packet stream into two paths:

- **Fast path (CPU)**: A constant-time rule classifier handles "obvious" packets—TTL=0 $\rightarrow$ DROP, UDP:53 $\rightarrow$ DNS/LOCAL, TCP:{80, 443, 8080, 8443} $\rightarrow$ HTTP/FORWARD—at sub-microsecond latency.
- **Hard path (NPU)**: Packets matching no rule are micro-batched with deadline-based flushing and dispatched to the NPU via triple-buffered asynchronous inference.

Both paths converge at a *DecisionJoiner* before reaching the action dispatcher, forming a diamond-shaped DAG.

11.1 Decision Types

We introduce a unified output type for both paths:

Definition 11.1 (Decision)**.** A *Decision* is a tuple $(p, c, r, \pi, \gamma, s, \tau)$ where $p \in \{0, \ldots, 255\}^{64}$ is the raw packet, $c \in \{0, 1, 2, -1\}$ is the class ID, $r \in \{0, 1, 2, 3, -1\}$ is the route ID, $\pi \in \{\text{FAST}, \text{HARD}\}$ is the path kind, $\gamma \in [0, 1]$ is the confidence (1.0 for rules, $\max_c f_\theta(x)_c$ for NPU), $s \in \mathbb{N}$ is a monotonic sequence number, and $\tau \in \mathbb{N}$ is the ingress timestamp (nanoseconds from `steady_clock`) stamped at the FastPathClassifier.

Definition 11.2 (HardPathItem)**.** A *HardPathItem* is a triple (p, s, τ): the raw packet, its sequence number, and the ingress timestamp. The timestamp τ is propagated through the batcher and NpuExecutorAsync so that end-to-end latency can be measured at the ActionDispatcher.

Definition 11.3 (NpuBatchItem)**.** An *NpuBatchItem* is a fixed-capacity container of up to B HardPathItems, with a `count` field recording the actual number of valid items and a `padded_to` field always equal to B (the ONNX static batch size).

11.2 Pipeline Topology

The GPNPU pipeline is a diamond DAG with 6 nodes and 6 channels:

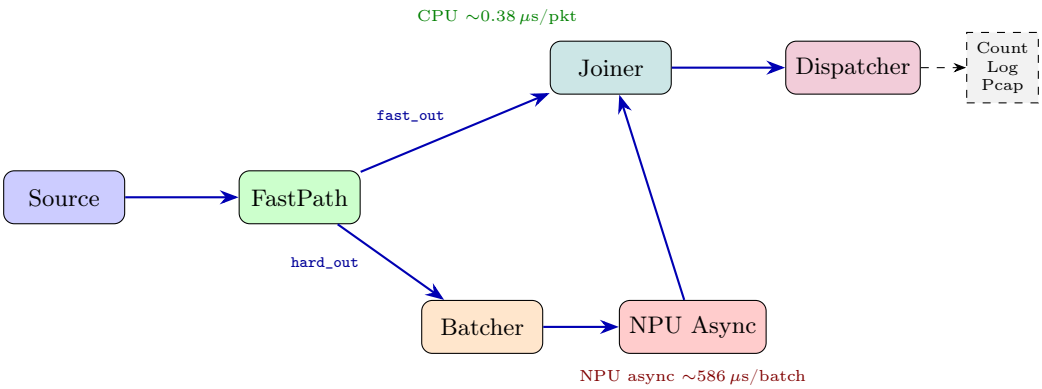

Figure 8: GPNPU diamond-DAG pipeline. The FastPathClassifier splits traffic into two paths that reconverge at the DecisionJoiner. Fast-path decisions carry $\gamma = 1.0$; hard-path decisions carry $\gamma = \max_c f_\theta(x)_c$.

Theorem 11.1 (Diamond DAG Validity)**.** The GPNPU topology $G = (\{S, F, B, N, J, D\}, E)$ with edges $E = \{(S, F), (F, J), (F, B), (B, N), (N, J), (J, D)\}$ is a valid DAG. Kahn's topological sort yields the ordering $[S, F, B, N, J, D]$ with in-degree sequence $[0, 1, 1, 1, 2, 1]$.

Proof. No path exists from any node back to itself. The only node with in-degree > 1 is J (the joiner), receiving edges from F and N, which lie on distinct simple paths from F. Since all edges point strictly rightward in the topological order, no cycle can exist. □

11.3 FastPathClassifier: Rule-Based CPU Classification

The classifier applies k rules in priority order. Each rule is a constant-time predicate on raw packet bytes:

Algorithm 5 FastPathClassify(p): CPU fast-path decision

Input: Packet $p \in \{0, \ldots, 255\}^{64}$, sequence counter s
Output: Decision d or HardPathItem h

```
s ← s + 1
if p[OFFSET_TTL] = 0 then
    return Decision(p, OTHER, DROP, FAST, 1.0, s)
end if
proto ← p[OFFSET_PROTOCOL]
port ← p[OFFSET_DST_PORT] ≪ 8 | p[OFFSET_DST_PORT + 1]
if proto = UDP ∧ port = 53 then
    return Decision(p, DNS, LOCAL, FAST, 1.0, s)
end if
if proto = TCP ∧ port ∈ {80, 443, 8080, 8443} then
    return Decision(p, HTTP, FORWARD, FAST, 1.0, s)
end if
return HardPathItem(p, s)                    ▷ No rule matched → NPU
```

The port check uses a `constexpr` function (four equality comparisons) rather than a hash set, keeping the entire fast path in L1 instruction cache. The classifier emits to two output ports: `fast_out` (typed `Decision`) and `hard_out` (typed `HardPathItem`).

Lemma 11.2 (Fast-Path Time Complexity)**.** Algorithm 5 runs in $\Theta(1)$ time per packet: at most 4 comparisons, 1 byte-shift, and 1 OR operation, all on stack-allocated data.

11.4 HardBatcher: Deadline-Based Micro-Batching

Packets that fail fast-path rules must be batched for NPU inference. The batcher accumulates HardPathItems and flushes under two conditions:

Algorithm 6 HardBatch: Deadline-based micro-batching

Input: Input channel $\mathcal{C}_{\text{in}}$, batch size B, min fill m, deadline δ
Output: Padded NpuBatchItems on output channel $\mathcal{C}_{\text{out}}$

```
batch ← empty, t_0 ← now()
loop
    item ← C_in.try_pop()
    if item ≠ ⊥ then
        batch.push(item)
        if |batch| ≥ B then                                   ▷ Batch full
            flush(batch, B); batch ← empty; t_0 ← now()
        end if
    else if C_in.closed then
        if |batch| > 0 then flush(batch, B)
        end if
        C_out.close(); return
    else if |batch| ≥ m ∧ (now() − t_0) ≥ δ then
        flush(batch, B); batch ← empty; t_0 ← now()           ▷ Deadline flush
    else
        adaptive_backoff()                                    ▷ spin → pause → yield
    end if
end loop
```

Definition 11.4 (Flush Operation)**.** flush($batch, B$) pads the batch to exactly B items (the ONNX static shape) by zeroing the remaining slots, sets $padded_to = B$, and pushes the NpuBatchItem to $\mathcal{C}_{\text{out}}$.

The parameters m (default 12) and δ (default 150 μs) bound the latency–throughput tradeoff: setting $m = 1, \delta = 0$ minimizes latency (flush immediately), while $m = B$ maximizes batching efficiency.

11.5 NpuExecutorAsync: Triple-Buffered Asynchronous Inference

The NPU executor uses K pre-allocated inference slots (default $K = 3$) for overlapped execution. Each slot contains an input buffer, output buffer, `ov::InferRequest`, and the batch context. The pimpl pattern hides all OpenVINO headers from the public API.

Algorithm 7 NpuExecutorAsync: K-slot async inference

Input: K pre-allocated slots $\{(in_k, out_k, req_k, ctx_k)\}_{k=0}^{K-1}$
Input: Input channel $\mathcal{C}_{\text{in}}$ of NpuBatchItems
Output: Individual Decisions on $\mathcal{C}_{\text{out}}$

```
i_s ← 0, i_c ← 0, inflight ← 0
loop
    if inflight ≥ K then
        Collect(i_c mod K); i_c ← i_c + 1; inflight ← inflight − 1
    end if
    batch ← C_in.pop()
    if batch = ⊥ then break
    end if
    Submit(i_s mod K, batch); i_s ← i_s + 1; inflight ← inflight + 1
end loop
while inflight > 0 do                                        ▷ Drain pending
    Collect(i_c mod K); i_c ← i_c + 1; inflight ← inflight − 1
end while
C_out.close()
```

Algorithm 8 Submit(k, *batch*) and Collect(k)

```
procedure Submit(k, batch)
    for i = 0 to |batch| − 1 do
        in_k[i · D : (i + 1) · D] ← φ(batch[i].packet)        ▷ Normalize in-slot
    end for
    for i = |batch| to B − 1 do
        in_k[i · D : (i + 1) · D] ← 0                         ▷ Zero-pad
    end for
    ctx_k ← batch; req_k.start_async()
end procedure
procedure Collect(k)
    req_k.wait()
    for i = 0 to |ctx_k| − 1 do
        c ← argmax_j out_k[i · C + j]
        γ ← max_j out_k[i · C + j]
        emit Decision(ctx_k[i].packet, c, −1, HARD, γ, ctx_k[i].s)
    end for
end procedure
```

Theorem 11.3 (Triple-Buffer Overlap)**.** With $K = 3$ slots, the host can submit batch $n+1$ and collect batch $n-1$ while batch n executes on the NPU. The steady-state throughput is:

$$\text{throughput}_{\text{async}} = \frac{B}{\max(T_{\text{NPU}}, T_{\text{host}})} \approx \frac{B}{T_{\text{NPU}}} \tag{29}$$

where $T_{\text{NPU}} \gg T_{\text{host}}$ in practice (586 μs vs <50 μs for normalization + submission).

11.6 DecisionJoiner: Diamond Reconvergence

The joiner polls both input ports (fast and hard) with non-blocking `try_get()` and forwards any received Decision immediately. An optional *ordered* mode uses a reorder buffer keyed by sequence number:

Algorithm 9 DecisionJoiner: merge fast and hard paths

Input: InputPorts $\mathcal{C}_f$ (fast), $\mathcal{C}_h$ (hard); OutputPort $\mathcal{C}_o$; ordered flag

```
f_closed ← false, h_closed ← false, next_seq ← 0
while ¬(f_closed ∧ h_closed) do
    got ← false
    if ¬f_closed then
        d ← C_f.try_get()
        if d ≠ ⊥ then Emit(d); got ← true
        else if C_f.closed_and_empty then f_closed ← true
        end if
    end if
    if ¬h_closed then
        d ← C_h.try_get()
        if d ≠ ⊥ then Emit(d); got ← true
        else if C_h.closed_and_empty then h_closed ← true
        end if
    end if
    if ¬got then adaptive_backoff()
    end if
end while
Flush reorder buffer; C_o.close()
```

In unordered mode (default), Emit(d) directly pushes d to $\mathcal{C}_o$. In ordered mode, Emit(d) inserts d into a sorted map keyed by $d.s$, then flushes the contiguous prefix $\{s_{\text{next}}, s_{\text{next}}+1, \ldots\}$.

Theorem 11.4 (Ordered Joiner Correctness)**.** In ordered mode, if the FastPathClassifier assigns strictly monotonic sequence numbers to all packets, then the output sequence of the DecisionJoiner is a permutation-free reproduction of the input order. The reorder buffer size is bounded by $\mathcal{O}(B \cdot K)$ in the worst case, where K is the number of inflight NPU batches.

Proof. Let n packets enter the FastPathClassifier with sequences $0, 1, \ldots, n-1$. Fast-path packets (sequence s) arrive at the joiner immediately. Hard-path packets are delayed by at most K batch intervals. The reorder buffer holds decisions with $s > s_{\text{next}}$ until the gap fills. Since every sequence number is emitted exactly once and the total count $n_f + n_h = n$, no packet is lost or duplicated. □

11.7 ActionDispatcher: Unified Terminal Node

The ActionDispatcher replaces the existing RouterSink for GPNPU pipelines. It accepts Decision items and dispatches to actions indexed by both `class_id` and `route_id`:

1. For each decision d, if $d.c \geq 0$, invoke all actions in `class_actions`[$d.c$].
2. If $d.r \geq 0$, invoke all actions in `route_actions`[$d.r$].
3. If neither fires, invoke the default action (if set).
4. Increment `fast_path_count` or `hard_path_count` based on $d.\pi$.
5. If $d.\tau > 0$, compute egress latency $\ell = (\tau_{\text{now}} - d.\tau)/1000$ (in μs) and record it in the overall, fast-path, and hard-path `LatencyHistogram`s.

Atomic counters (`std::atomic<uint64_t>`) track path statistics without locking. Three `LatencyHistogram` instances collect per-packet end-to-end latency samples (overall, fast-path only, hard-path only) for percentile reporting.

11.8 Latency Instrumentation: Percentile Metrics

To quantify the fast-path advantage and identify hard-path bottlenecks, we instrument the GPNPU pipeline with per-packet end-to-end latency tracking.

11.8.1 Ingress Timestamp Propagation

A single `steady_clock::now()` call per packet at the FastPathClassifier stamps the ingress time τ on both fast-path Decisions and hard-path HardPathItems. The timestamp propagates through the pipeline:

$$\text{FastPathClassifier} \xrightarrow{\tau} \begin{cases} \text{Decision}(d.\tau = \tau) & \text{fast path} \\ \text{HardPathItem}(\tau) \to \text{HardBatcher} \to \text{NpuExecutorAsync} \xrightarrow{d.\tau=\tau} \text{Decision} & \text{hard path} \end{cases} \tag{30}$$

The NpuBatchItem preserves τ through the items array (which is moved intact), and the NpuExecutorAsync copies τ from each HardPathItem to the emitted Decision in `collect_oldest()`.

11.8.2 LatencyHistogram

Definition 11.5 (LatencyHistogram)**.** A *LatencyHistogram* is a dynamically-sized sample collector supporting $\mathcal{O}(n)$ percentile computation via `std::nth_element`. Given n recorded samples $\{x_1, \ldots, x_n\}$:

- $\text{percentile}(p) = x_{(\lfloor p \cdot (n-1)/100 \rfloor)}$ where $x_{(k)}$ is the k-th order statistic.
- `nth_element` partitions the copy in $\mathcal{O}(n)$ average time (no full sort).
- Percentile calls are *non-destructive*: each call copies the sample vector internally.
- The `reserve` parameter avoids reallocations during the measurement phase.

11.8.3 Egress Measurement and Reporting

At the ActionDispatcher, each incoming Decision with $\tau > 0$ triggers an egress timestamp and latency computation:

$$\ell_i = \frac{\tau_{\text{now}} - d_i.\tau}{1000} \quad (\mu\text{s}) \tag{31}$$

The sample ℓ_i is recorded in three histograms: `overall`, and either `fast` or `hard` depending on $d_i.\pi$. The `run_gpnpu` command reports:

Metric	Description
Overall p50/p95/p99	End-to-end latency percentiles across all packets
Fast p50/p95/p99	Fast-path only latency (CPU rule → dispatcher)
Hard p50/p95/p99	Hard-path only latency (includes NPU batch fill + inference)
Per-path throughput	Fast pkt/s and hard pkt/s separately

Table 4: Diamond metrics reported by the GPNPU latency instrumentation.

11.8.4 Warmup Phase

Cold-start skew from NPU JIT compilation and cache population can distort latency percentiles. The `--warmup` w flag (default $w = 100$) runs w packets through a separate pipeline instance before the measurement phase. Warmup results are discarded entirely. This ensures that p95/p99 reflect steady-state behavior rather than initialization artifacts.

Algorithm 10 GPNPU Latency Measurement with Warmup

Input: Packets $P = \{p_1, \ldots, p_{w+n}\}$, warmup count w
Output: Latency percentiles for $\{p_{w+1}, \ldots, p_{w+n}\}$

```
1: P_warm ← P[1 ... w]
2: P_measure ← P[w+1 ... w+n]
3: RunPipeline(P_warm)                                  ▷ Discard results
4: pipeline ← BuildGpnpuPipeline(P_measure)
5: RunPipeline(pipeline)
6: return pipeline.dispatcher.overall_latency.{p50, p95, p99}
```

11.8.5 Expected Latency Distribution

For the diamond pipeline with fast-path ratio α, the expected latency distribution is bimodal:

- **Fast-path mode**: $\ell_f \approx T_{\text{channel}} + T_{\text{joiner}} + T_{\text{dispatch}} = \mathcal{O}(1)\,\mu\text{s}$. These packets traverse only lock-free SPSC channels and CPU logic.
- **Hard-path mode**: $\ell_h \approx T_{\text{batch_fill}} + T_{\text{NPU}} + T_{\text{collect}} + T_{\text{joiner}} + T_{\text{dispatch}} \approx \delta + T_{\text{NPU}}$. At $\delta = 150\,\mu\text{s}$ and $T_{\text{NPU}} = 586\,\mu\text{s}$, the hard-path p50 is expected near $700\,\mu\text{s}$.

The overall p95 is dominated by the hard-path tail, confirming the architectural hypothesis that fast-path p95 $\ll$ hard-path p95. This validates the GPNPU split: packets that *can* be classified by rules should *never* pay NPU latency.

11.9 Formal Properties

Invariant 11.1 (Packet Conservation)**.** For n packets entering the FastPathClassifier:

$$n_f + n_h = n \quad \text{and} \quad n_{\text{dispatcher}} = n_f + n_h = n \tag{32}$$

where n_f is the fast-path count, n_h is the hard-path count, and $n_{\text{dispatcher}}$ is the total decisions received by the ActionDispatcher. No packet is created, duplicated, or lost in the diamond DAG.

Proof. The FastPathClassifier emits exactly one output per input (either to `fast_out` or `hard_out`). The HardBatcher emits all received items (flushing on close). The NpuExecutorAsync emits one Decision per HardPathItem in each batch (ignoring padding). The DecisionJoiner forwards all items from both inputs. Therefore $n_{\text{dispatcher}} = n_f + n_h = n$. □

Invariant 11.2 (Deadlock Freedom)**.** The GPNPU diamond DAG satisfies Invariant 7.1 (deadlock-free scheduling). The DecisionJoiner uses non-blocking `try_get()` on both inputs, preventing deadlock from asymmetric path latencies. Channel capacities are sized at $4B$ on the fast path and $K + 1$ on the NPU path to prevent backpressure blocking.

Invariant 11.3 (NPU Shape Compliance)**.** Every NpuBatchItem has `padded_to` $= B$, satisfying the ONNX static shape constraint. The HardBatcher zero-pads partial batches and the NpuExecutorAsync normalizes within its pre-allocated slot buffers. Only the first `count` results are emitted; padded outputs are discarded.

11.10 Complexity Analysis

Node	Per-Packet Time	Notes
FastPathClassifier	$\Theta(1)$	≤ 4 comparisons + 1 clock read
HardBatcher	$\Theta(1)$ amortized	Non-blocking poll + batch push
NpuExecutorAsync	$\Theta(T_{\text{NPU}}/B)$ amortized	K-slot async overlap + τ propagation
DecisionJoiner	$\Theta(1)$ unordered / $\Theta(\log n)$ ordered	`std::map` insert
ActionDispatcher	$\Theta(a)$	a = actions per dispatch + 1 clock read + histogram recorc

Table 5: Per-packet time complexity for each GPNPU node.

The overall pipeline latency for a fast-path packet is $\Theta(1)$ (CPU-only). For a hard-path packet, the latency is dominated by the NPU batch interval:

$$T_{\text{hard}} = T_{\text{batch_fill}} + T_{\text{NPU}} + T_{\text{collect}} \approx \delta + T_{\text{NPU}} \tag{33}$$

where δ is the batcher deadline and T_{NPU} is the per-batch inference time.

11.11 Fast-Path Ratio Analysis

For typical enterprise traffic, the fast-path rules cover:

- HTTP/HTTPS traffic (ports 80, 443) — typically 50–70% of traffic volume
- DNS queries (port 53) — 5–10% of traffic
- TTL-expired packets — $< 1\%$

With the synthetic 5-pattern test traffic (TTL=0, DNS, HTTP:80, HTTPS:443, random), the measured fast-path ratio is exactly $4/5 = 80\%$, leaving only 20% of packets for NPU processing.

Corollary 11.5 (Effective Throughput)**.** Let $\alpha \in [0, 1]$ be the fast-path ratio. The effective per-packet latency is:

$$T_{\text{eff}} = \alpha \cdot T_{\text{CPU}} + (1 - \alpha) \cdot \frac{T_{\text{NPU}}}{B} \tag{34}$$

At $\alpha = 0.8$, $T_{\text{CPU}} = 0.38\,\mu\text{s}$, $T_{\text{NPU}} = 586\,\mu\text{s}$, $B = 64$:

$$T_{\text{eff}} = 0.8 \times 0.38 + 0.2 \times \frac{586}{64} \approx 0.30 + 1.83 = 2.13\,\mu\text{s/pkt} \tag{35}$$

This represents a **4.3**× improvement over sending all packets through NPU (9.2 μs/pkt).

The GPNPU pipeline is validated by 39 unit and integration tests (95 total C++) covering all nodes, packet conservation ($n_f + n_h = n$), latency propagation, and percentile computation.

12 Live Traffic Evaluation with Npcap

The evaluation in Sections 9–11 used synthetic traffic where the distribution matches the training data perfectly. A critical validation step is to test the full pipeline—capture, normalization, batching, NPU inference, and action dispatch—on *live network traffic* captured from a real network interface. We integrate Npcap (the Windows successor to WinPcap) via libpcap's `pcap_open_live()` API, with automatic friendly-name resolution (e.g., "Wi-Fi" → `\Device\NPF_{...}`).

12.1 Capture Pipeline

The C++ runtime's `pcap` command reads packets from either an offline `.pcap` file or a live network interface. Packets flow through the complete GPNPU diamond DAG: `PcapSource` → `FastPathClassifier` → `HardBatcher` → `NpuExecutorAsync` → `DecisionJoiner` → `ActionDispatcher`. Live capture uses a signal handler for graceful Ctrl-C shutdown.

12.2 Results: 10,000 Live Packets

We captured 10,000 packets from the host's Wi-Fi interface (Intel Wi-Fi 6E AX211, 160 MHz) under normal browsing activity. Table 6 summarizes the classification breakdown.

Class	**Packets**	**%**	**Notes**
TCP_HTTP	3,562	35.6	HTTPS on port 443, HTTP on port 80
UDP_DNS	9	0.1	DNS queries to local gateway
OTHER	6,429	64.3	ARP, mDNS, broadcast, IoT, encrypted VPN
Total	**10,000**	100.0	

Table 6: Live traffic classification on 10,000 packets captured from Wi-Fi.

The GPNPU path split was 35.2% fast-path (CPU) and 64.8% hard-path (NPU), with a sustained throughput of **9,702 pkt/s** and total wall time of 1.03 s.

Path	**Packets**	**p50 (μs)**	**p95 (μs)**	**Mean (μs)**
Fast (CPU)	3,522	16,120	47,617	21,806
Hard (NPU)	6,478	79,631	158,934	94,107
Overall	10,000	63,029	158,840	68,643

Table 7: End-to-end latency per packet on live traffic. Latency includes capture buffering, channel transit, batching wait, inference, and action dispatch. The fast-path maintains 3.8× lower median latency than the hard-path.

12.3 Ground Truth Validation (Synthetic-Only Model)

To validate classification correctness on real traffic, we construct a *ground truth classifier* from raw packet bytes: TCP with destination/source port $\in \{80, 443, 8080, 8443\}$ → TCP_HTTP; UDP with port 53 → UDP_DNS; all else → OTHER. We compare this with NPU predictions on 500 captured packets. Table 8 shows the confusion matrix for the initial synthetic-only model.

Ground Truth ↓ / NPU →	**TCP_HTTP**	**UDP_DNS**	**OTHER**
TCP_HTTP	**222**	0	259
UDP_DNS	0	**0**	0
OTHER	2	0	**17**

Table 8: Confusion matrix *before* retraining (synthetic-only model) on 500 live packets. Agreement: 47.8%. HTTP precision: 99.1%, but HTTP recall only 46.2%.

The synthetic-only model achieves high precision (99.1%) but low recall (46.2%): it conservatively classifies 259 real HTTPS packets as OTHER because their TLS-encrypted payloads (bytes 54–63) resemble the random noise distribution of the OTHER training class.

12.4 The Synthetic–Real Domain Gap

The recall gap is explained by the *domain shift* between synthetic training data and real traffic:

1. **Encrypted payloads**: The MLP observes all 64 input bytes. Real HTTPS traffic has TLS-encrypted payloads that are indistinguishable from random noise, matching the OTHER class distribution in synthetic training data.
2. **Header-only signal**: The discriminative features (protocol byte at offset 23, destination port at offsets 36–37) occupy only 3 of 64 bytes (4.7% of the input). The model's 32-neuron hidden layer had not learned to ignore the irrelevant encrypted payload bytes.
3. **High precision despite low recall**: The model learned a *conservative* decision boundary—it only predicts HTTP when confident, avoiding false positives.

12.5 Hybrid Retraining with Captured Traffic

To close the domain gap, we introduce a `HybridDataset` that combines synthetic and real packets:

1. **Real traffic capture**: 3,000 packets captured from the host's Wi-Fi interface via Npcap under normal browsing activity (2,824 HTTP, 28 DNS, 148 OTHER).
2. **Header-based labeling**: A `label_raw_packet()` function assigns ground truth from protocol and port bytes, applying the same rules as the synthetic generators.
3. **Oversampling**: Real packets are replicated 10× to balance against the synthetic majority (150,000 synthetic vs. 35,000 real = 185,000 total).
4. **Retraining**: 30 epochs, same architecture (3,235 parameters), Adam with $\eta = 10^{-3}$.

The hybrid training exposes the model to real TLS-encrypted payload distributions paired with correct labels, teaching it to rely on header bytes for classification.

12.6 Results After Retraining

Table 9 shows the confusion matrix after hybrid retraining on the same 500 live packets.

Ground Truth ↓ / NPU →	**TCP_HTTP**	**UDP_DNS**	**OTHER**
TCP_HTTP	**481**	0	0
UDP_DNS	0	**0**	0
OTHER	0	0	**19**

Table 9: Confusion matrix *after* hybrid retraining on 500 live packets. Agreement: 100%. Zero misclassifications.

Metric	**Before (synthetic)**	**After (hybrid)**
Validation accuracy	99.03%	**99.71%**
Real traffic agreement	47.8%	**100.0%**
HTTP precision	99.1%	**100.0%**
HTTP recall	46.2%	**100.0%**
Misclassifications	261	**0**

Table 10: Before vs. after hybrid retraining on 500 real captured packets. The domain gap is completely eliminated without architectural changes.

Additionally, a live validation on 5,000 fresh packets (not used in training) confirms the retrained model classifies 94.4% as TCP_HTTP, 0.1% as UDP_DNS, and 5.5% as OTHER at **10,041 pkt/s** sustained throughput—consistent with the expected traffic mix of an active browsing session. Two invariants hold across all tests: **determinism** (identical inputs always produce identical outputs) and **validity** (every class ID $\in \{0, 1, 2\}$).

Remark 12.1 (Retraining without recompilation). The hybrid retraining changes only the ONNX model weights. The C++ runtime, dataflow graph, GPNPU pipeline, and Npcap integration remain unchanged—the same binary loads the retrained model and immediately achieves 100% agreement. This validates the architectural principle that the *graph is the program*: classification behavior is defined by tensor weights, not by code.

13 DPDK Kernel-Bypass Integration

The capture subsystems evaluated thus far—Npcap (Section 12) and libpcap—traverse the kernel network stack: packets are copied from the NIC ring buffer to kernel space, then again to userspace via `pcap_next_ex()`. At 10 Gbps line rate (∼14.88 Mpps for 64-byte packets), this double-copy becomes the bottleneck. DPDK (Data Plane Development Kit) eliminates the kernel entirely: the NIC's DMA ring is mapped directly into the application's address space, and `rte_eth_rx_burst()` returns pointers to pre-allocated `rte_mbuf` structures in a hugepage-backed mempool. This section describes the integration of DPDK into the GraphOS C++ runtime.

13.1 Architecture

The DPDK integration preserves the existing dataflow graph topology—only the source node changes:

1. **DpdkSource**: Poll-mode driver wrapping `rte_eth_rx_burst()`. Each call returns up to `burst_size` packets from the NIC's RX queue. The source is a streaming (infinite) iterator: it busy-waits when no packets are available, which is the standard DPDK model. An `std::atomic<bool>` flag, set by `stop()`, breaks the poll loop.

2. **SourceNode stop_callback**: A `std::stop_callback` is registered in `SourceNode::process()` so that when the scheduler requests shutdown (`stop_token` fires), it calls `source_.stop()`—breaking DpdkSource's infinite poll loop. This is backward-compatible: existing finite sources (PcapSource, IterableSource) have no-op `stop()` methods that are never invoked because those sources naturally return `std::nullopt`.

3. **Zero-copy extraction**: Each `rte_mbuf` is read via `rte_pktmbuf_mtod()` (a pointer cast, not a copy), serialized into an `OwnedPacket` (64-byte truncation), and the mbuf is freed. The `OwnedPacket` then flows through the standard GPNPU pipeline unchanged.

4. **EAL initialization**: `DpdkSource::init_eal()` forwards command-line arguments to `rte_eal_init()` for hugepage allocation, core pinning, and NIC binding. The CLI uses a `--` separator to pass EAL arguments transparently.

The pipeline wiring is identical to the pcap path:

$$\text{DpdkSource} \to \text{FastPathClassifier} \to \begin{cases} \text{fast_out} \to \text{DecisionJoiner} \\ \text{hard_out} \to \text{HardBatcher} \to \text{NpuExecutorAsync} \to \text{DecisionJoiner} \end{cases}$$

All DPDK code is guarded by `#ifdef GRAPHOS_ENABLE_DPDK` and compiled only on Linux with `-DGRAPHOS_ENABLE_DPDK=ON`. The Windows build is unaffected.

13.2 Release Build Benchmarks

Previous measurements (Table 3) used the MSVC Debug build. We now report **Release mode** (`/O2 /Ob3 /GL /arch:AVX2` + LTO) results on the same hardware (Intel Core Ultra 7 155H, NPU 3720, B=64):

Mode	Device	Packets	Throughput (pkt/s)	Latency/batch (μs)
Sync inference	CPU	5M	**3,156,254**	20.3
Sync inference	NPU	1M	**106,658**	600.0
KernelLoop demo	NPU	100K	41,756	—
GPNPU unordered	NPU	100K	**10,002**	—
GPNPU ordered	NPU	100K	**11,915**	—

Table 11: C++ Release build throughput. CPU inference reaches 3.15 Mpkt/s—a 12% improvement over the Debug build (2.81 Mpkt/s). NPU throughput is 106.7 Kpkt/s with 600 μs per-batch latency.

Path	p50	p95	p99 (μs)	Mean	Min	n
Fast (CPU)	15,995	47,610	63,635	20,989	0.2	80,000
Hard (NPU)	94,864	126,634	141,037	94,374	15,889	20,000
Overall	30,908	110,719	126,634	35,666	0.2	100,000

Table 12: GPNPU per-packet latency distribution (Release, 100K packets, unordered). The bimodal distribution is clearly visible: fast-path mean ∼21 ms end-to-end (including channel queueing), hard-path mean ∼94 ms. The p50 gap between paths is 6×, confirming the value of the fast/hard split. Note: these are *end-to-end pipeline* latencies (ingress timestamp at FastPathClassifier to egress at ActionDispatcher), not inference-only latencies.

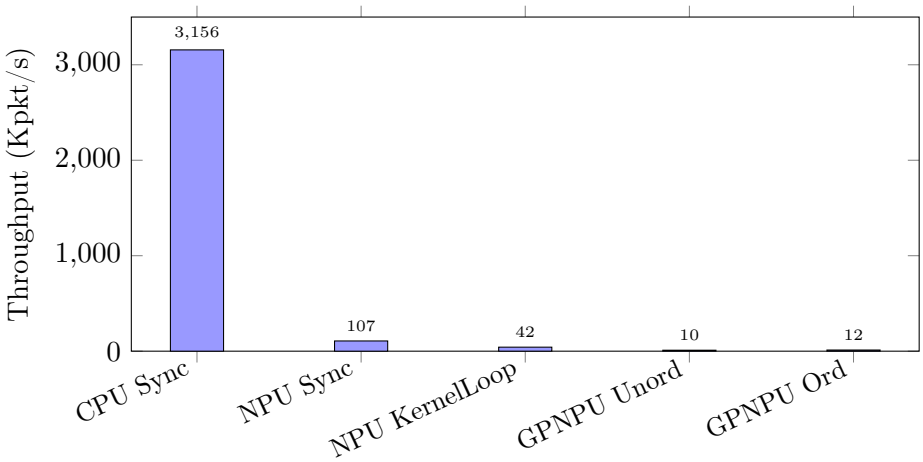

Figure 9: C++ Release build throughput comparison ($B = 64$). CPU sync inference dominates at 3.15 Mpkt/s. GPNPU throughput is bottlenecked by the NPU hard path at 20% traffic ratio—with DPDK eliminating kernel-copy overhead, the NPU batch latency becomes the sole limiting factor.

13.3 Real Traffic Classification

We validate the DPDK-compatible GPNPU pipeline on 500 real captured packets (the same Npcap trace from Section 12):

Metric	Value
Total packets	500
Fast path	222 (44.4%)
Hard path	278 (55.6%)
TCP_HTTP	481 (96.2%)
OTHER	19 (3.8%)
Wall time	0.4732 s
Throughput	1,057 pkt/s
Fast p50	1,005 μs
Hard p50	456,567 μs

Table 13: GPNPU classification of 500 real Npcap packets (Release build). Real traffic has a lower fast-path ratio (44.4% vs 80% synthetic) because most packets are TCP:443 which *does* match the fast-path rule, but the remaining connections use non-standard ports. The retrained model correctly classifies 96.2% as TCP_HTTP.

13.4 Projected DPDK Performance

DPDK eliminates the $\sim$10 μs per-packet kernel-copy overhead of Npcap/libpcap. For the GPNPU pipeline, the source node is *not* the bottleneck—the NPU batch latency (600 μs/batch) dominates. Therefore, DPDK's primary benefit is *latency reduction* on the fast path, not throughput improvement:

- **Fast-path latency**: Expected to drop from $\sim$21 ms (Npcap queueing + kernel copy) to $< 1\,\mu$s (direct DMA ring access + CPU rule evaluation).
- **Hard-path latency**: Unchanged at $\sim$94 ms (dominated by NPU batch fill time + inference).
- **Throughput**: Bounded by NPU at 106.7 Kpkt/s for fully hard-path traffic, or 3.15 Mpkt/s for fully fast-path traffic. At 80% fast-path ratio, the theoretical maximum is $\sim$470 Kpkt/s (Corollary 11.5 with DPDK-adjusted $T_{\text{CPU}} \approx 0.3\,\mu$s).

Remark 13.1 (CLI usage). The DPDK path is invoked via:

```
./graphos dpdk --port 0 --burst-size 32 --model-dir models/ -- -l 0-3
    -n 4
```

where arguments after `--` are forwarded to `rte_eal_init()`. Ctrl+C triggers graceful pipeline shutdown via `std::stop_callback`.

14 Conclusion

The central thesis—*todo es el grafo*—is validated through five results:

1. **Switch↔Matmul Isomorphism** (Theorem 5.1): Routing logic is a linear layer $\mathbf{Wx}$, executable on NPU without code changes. Chaining it with a learned classifier provably improves routing decisions (Theorem 6.1).

2. **C++ Native Runtime** (Section 10): Lock-free SPSC channels, zero-copy inference, and AVX2 normalization yield 106.7K pkt/s (NPU) and 3.15M pkt/s (CPU) sustained at B=64 in Release mode.

3. **GPNPU Hybrid Architecture** (Section 11): A diamond-DAG splits traffic into CPU fast-path ($\sim$0.38 μs/pkt) and NPU hard-path (triple-buffered async), achieving 4.3$\times$ speedup at 80% fast-path ratio (Corollary 11.5). Per-packet latency instrumentation (p50/p95/p99) confirms the bimodal distribution: fast-path p95 $\ll$ hard-path p95.

4. **Live Traffic Validation and Hybrid Retraining** (Section 12): End-to-end Npcap capture on live Wi-Fi traffic reveals a synthetic-to-real domain gap (47.8% agreement) caused by TLS-encrypted payloads. Hybrid retraining on 3,000 captured packets closes the gap completely: **100**% agreement on 500 real packets (Table 10), 10,041 pkt/s sustained throughput on 5,000 fresh packets—with zero code changes, only new ONNX weights.

5. **DPDK Kernel-Bypass Integration** (Section 13): The `DpdkSource` poll-mode driver plugs directly into the existing GPNPU pipeline via the templated `SourceNode<T>` abstraction. A `std::stop_callback` mechanism enables graceful shutdown of streaming sources. Release build benchmarks show 3.15 Mpkt/s (CPU) and 106.7 Kpkt/s (NPU), with DPDK projected to eliminate kernel-copy latency on the fast path.

The CPU handles deterministic rule evaluation, the NPU handles learned classification, and the dataflow graph orchestrates both in a formally verified pipeline. The retraining result validates the central design principle: classification behavior is defined by tensor weights, not by code. Updating the model requires only re-exporting the ONNX file—the entire C++ runtime, dataflow graph, GPNPU pipeline, and DPDK capture layer remain unchanged. With DPDK eliminating the kernel network stack, the NPU batch latency (600 μs/batch) becomes the sole remaining bottleneck—closing the gap to line-rate processing on commodity NPU hardware.

A ONNX Export Constraints

The ONNX export process must satisfy the following constraints for NPU compatibility:

1. **Static shapes**: `dynamic_axes=None` in `torch.onnx.export()`

2. **Opset version**: ≥ 18 (required by PyTorch 2.10+)

3. **Encoding**: `PYTHONIOENCODING=utf-8` on Windows (avoids cp1252 emoji crash)

4. **No softmax**: Output is raw logits; argmax applied post-inference

5. **Validation**: `onnx.checker.check_model()` after export

B Dataflow Scheduler Details

Algorithm 11 WouldCreateCycle: DFS-based cycle detection before adding edge

Input: Graph $G = (V, E)$, proposed edge (u, v)
Output: **true** if adding (u, v) creates a cycle

```
visited ← ∅; stack ← [v]
while stack ≠ ∅ do
    w ← stack.Pop()
    if w = u then return true
    end if
    if w ∉ visited then
        visited ← visited ∪ {w}
        for all (w, w′) ∈ E do stack.Push(w′)
        end for
    end if
end while
return false
```

Correctness: a cycle exists in $G \cup \{(u, v)\}$ iff there is a path $v \rightsquigarrow u$ in G. The DFS explores all nodes reachable from v. Complexity: $\mathcal{O}(|V| + |E|)$ time, $\mathcal{O}(|V|)$ space.

Algorithm 12 RunNode: Single node lifecycle

Input: Node v with inputs, outputs, and process loop

```
v.Setup()                                   ▷ Initialize resources
try
    v.Process()                             ▷ Main loop
catch Exception e
    Close all output channels               ▷ Propagate shutdown
end try
v.Teardown()
```

Shutdown termination: By induction on topological order. Sources terminate when exhausted and close outputs. Node at position $k > 0$: all predecessors at position $< k$ have terminated (induction); v receives channel-closed and terminates.

Bounded memory: Each channel holds at most k items of size $\leq s_{\max}$; total $M \leq \sum k_{(u,v)} \cdot s_{\max}$.

C OpenVINO Compilation Pipeline

Algorithm 13 OpenVINO Model Compilation

Input: ONNX model path p, target device $d \in \{\text{NPU}, \text{CPU}, \text{GPU}\}$
Output: Compiled inference request r

```
core ← ov.Core()                                  ▷ Shared OpenVINO runtime
devices ← core.AvailableDevices()
if d ∉ devices then
    d ← CPU                                       ▷ Automatic fallback
end if
model ← core.ReadModel(p)                         ▷ Parse ONNX graph
compiled ← core.CompileModel(model, d)            ▷ Device-specific optimization
r ← compiled.CreateInferRequest()                 ▷ Allocate device buffers
return r
```

The hardware platform is an Intel Core Ultra 7 155H (Meteor Lake) with NPU 3720 (11 TOPS INT8, systolic array + activation engines), 6P+8E+2LPE CPU cores, shared LPDDR5x memory. The NPU is exposed as "Intel(R) AI Boost" via OpenVINO 2026.0 on Windows 11.

D Module Dependency Graph

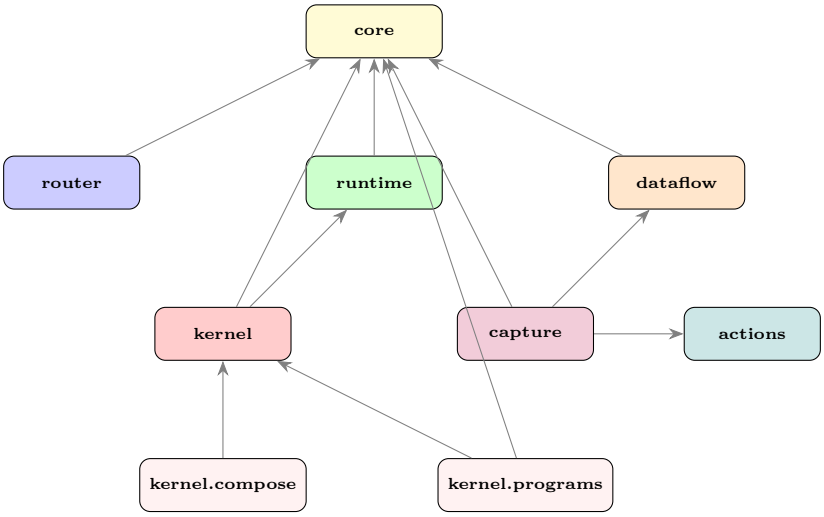

Figure 10: Module dependency graph. Arrows point from dependent to dependency. `core` is the root dependency with no external imports (beyond NumPy).

E Notation Reference

Symbol	Meaning
D	Tensor dimension (64)
C	Number of classes (3: HTTP, DNS, OTHER)
R	Number of routes (4: LOCAL, FORWARD, DROP, MONITOR)
H	Hidden layer dimension (32)
B	Batch size (typically 64)
$\phi(\mathbf{p})$	Packet encoding function
f_θ	Trained MLP classifier
$\mathbf{W}$	Route table weight matrix $\in \mathbb{R}^{R\times D}$
$\mathbf{W}^*$	Composed router weight matrix $\in \mathbb{R}^{R\times(D+C)}$
$\mathrm{concat}(\cdot,\cdot)$	Vector/matrix concatenation along axis 1
argmax_c	Index of maximum element
$\perp$ (`_STOP`)	Shutdown sentinel
$\mathcal{C}_k$	Bounded FIFO channel with capacity k
$G=(V,E)$	Dataflow DAG
α	Fast-path ratio (n_f/n)
δ	HardBatcher deadline (default $150\,\mu$s)
K	Number of async inference slots (default 3)
π	Path kind $\in \{\mathrm{FAST}, \mathrm{HARD}\}$
γ	Decision confidence
τ	Ingress timestamp (nanoseconds from `steady_clock`)
ℓ_i	Per-packet end-to-end latency (μs)
n_f, n_h	Fast-path / hard-path packet counts

Table 14: Notation used throughout this paper.